MINI PROJECT | FINAL EVALUATION

# Real-Time Network Intrusion Detection System

using Packet Inspection and Web-Based Alerting

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# Problem Statement

## Small networks have no real-time way to see attacks as they happen.

**Where:** homes, hostels and college labs

**Today:** attacks pass unseen in the packets

**Gap:** tools are manual, complex or unexplainable

**Need:** light, explainable, alerts within a second

ATTACKS DETECTED

### Port Scan

tries every port to find an open one

### SYN Flood

floods a server with half-open connections

### ARP Spoofing

puts the attacker in the middle of traffic

# Objectives: All Achieved

- ✓ Capture live TCP, UDP and ARP packets
- ✓ Detect port scans with a sliding-window rule
- ✓ Detect SYN floods from half-open handshakes
- ✓ Detect ARP spoofing from a trust table
- ✓ Alert pipeline to a Node.js and Express server
- ✓ Live web dashboard, updated without refresh
- ✓ Automated tests and measured performance

**All seven objectives were met.** All three attack rules are implemented, tested and shown live on the dashboard.

# Existing Systems and the Gap

| Existing tool | What it does well | Gap for small networks |
|---|---|---|
| **Wireshark** | Captures and decodes every packet | Manual study, no automatic alerts |
| **Snort, Suricata** | Professional rule-based detection | Complex rules, expert setup, log files |
| **Zeek** | Deep analysis with scripts | Steep learning curve, no live dashboard |
| **ML-based IDS** | High accuracy on test datasets | Black box, needs training data |

MY SYSTEM FILLS THE GAP

**Lightweight**

about 200 lines, runs on a laptop

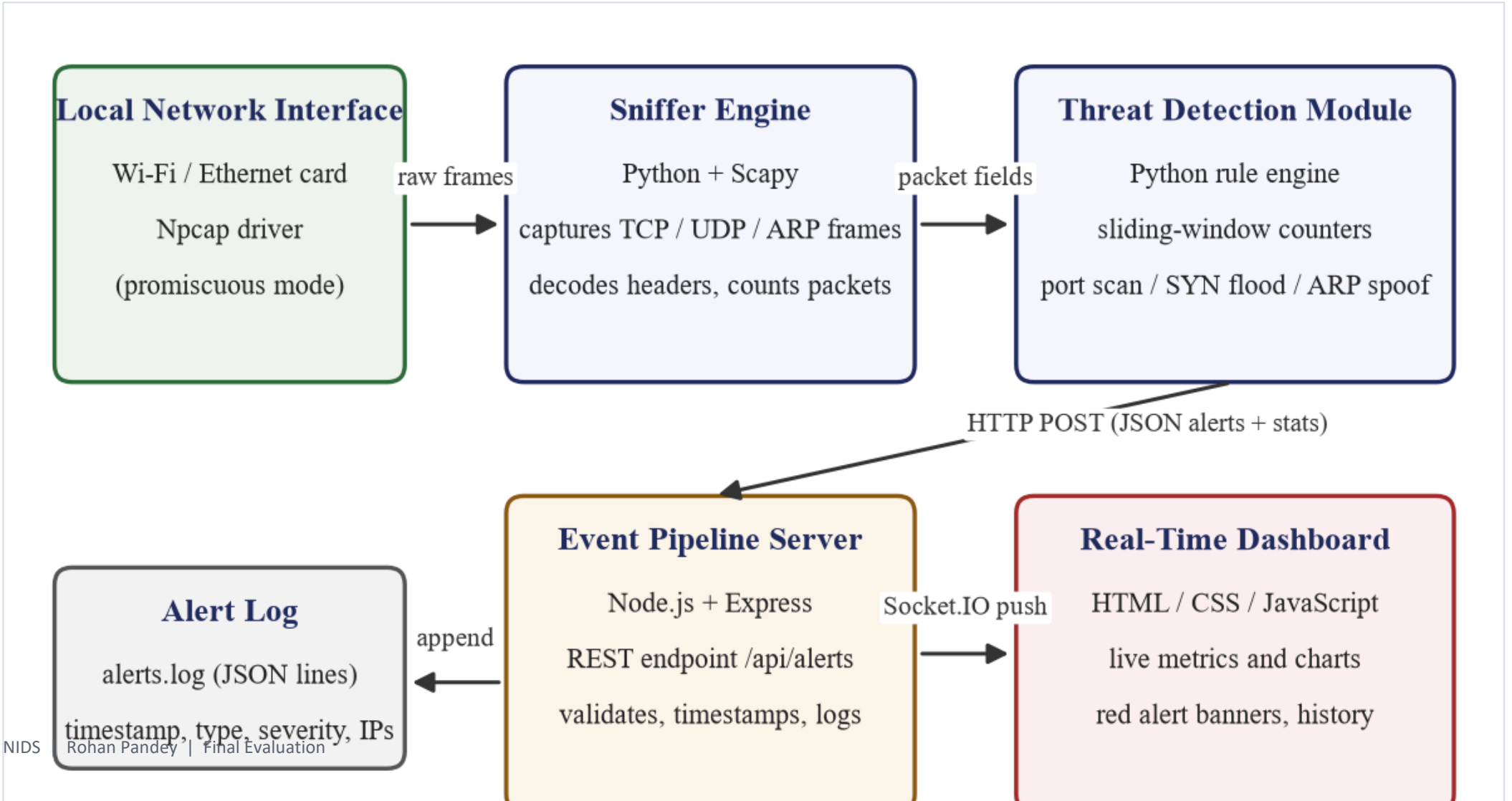
**Explainable**

every alert shows its threshold

**Live**

alert on the dashboard in under 1 second

# System Architecture



# Technology Stack

CAPTURE

## Npcap 1.89

raw packets on Windows

SENSOR

## Python 3.13 + Scapy 2.8

decodes IP, TCP, UDP, ARP

RULES

## Python standard library

deque window, set of ports

SERVER

## Node.js 24 + Express 5

receives alerts by HTTP POST

LIVE PUSH

## Socket.IO 4

alerts reach the browser instantly

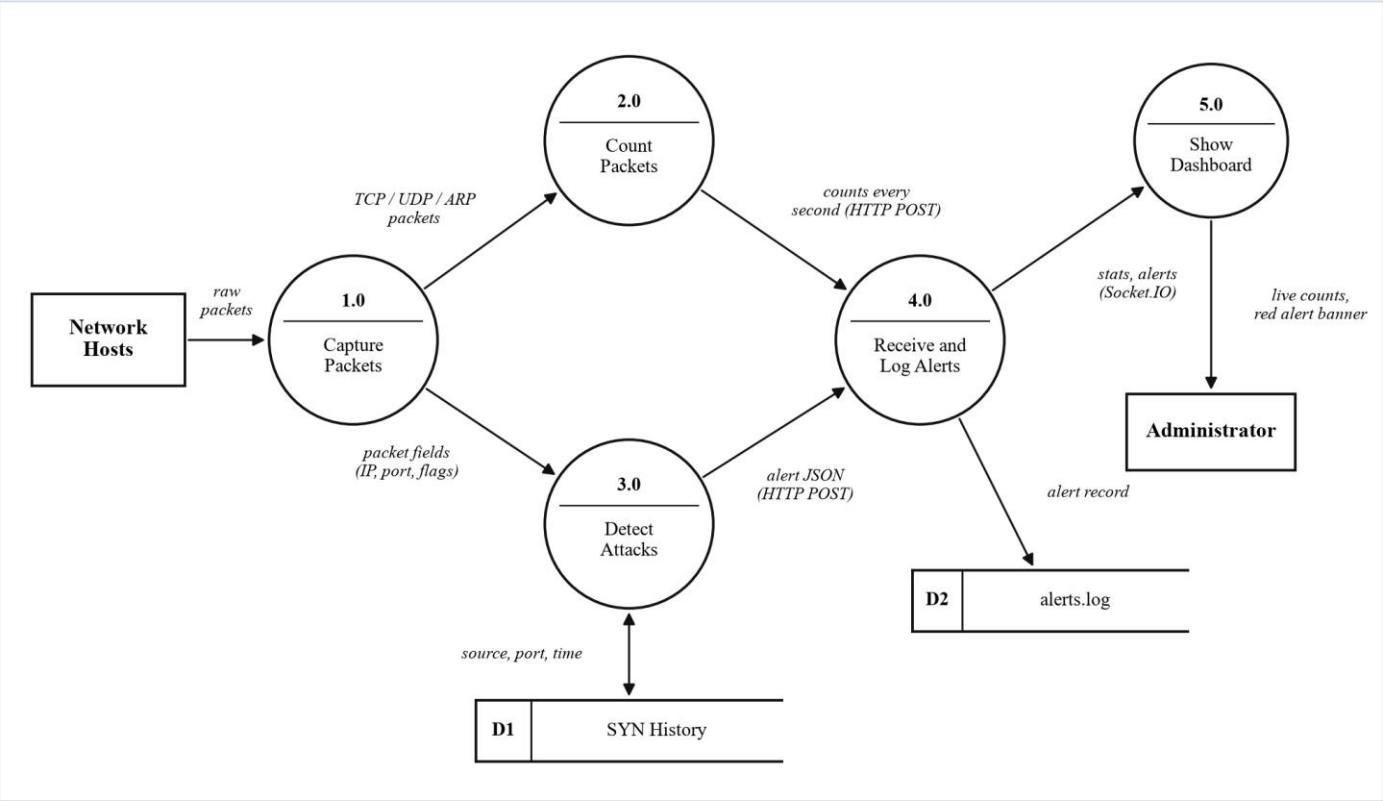
DASHBOARD

## HTML, CSS, JS + Chart.js

live chart, table and banner

**Storage:** JSON alert log **Testing:** unittest, Nmap, Wireshark **Tools:** VS Code, Git, GitHub

# System Design




Level-1 Data Flow Diagram

**DESIGN ARTEFACTS**

- Context and level-1 DFDs
- Use-case diagram
- Sequence diagram
- Component diagram
- Deployment diagram
- Flowchart of the port-scan rule

All diagrams are in the report, **numbered chapter-wise** as Fig. 3.1 to 4.2.

# Detection Rules

## Port Scan

> 15 distinct ports

from one source

within 5 seconds

**medium**

## SYN Flood

100+ half-open SYNs

to one host

few completed

**high**

## ARP Spoofing

gateway IP seen

with a different

MAC address

**critical**

**All rules are deterministic.** Each keeps a few seconds of history in a sliding window, compares a count against a threshold, and raises an alert with a cool-down so one attack gives one alert.

# Testing

**UNIT TESTS**

**10 / 10**

tests passing
one per rule, with
positive and negative cases

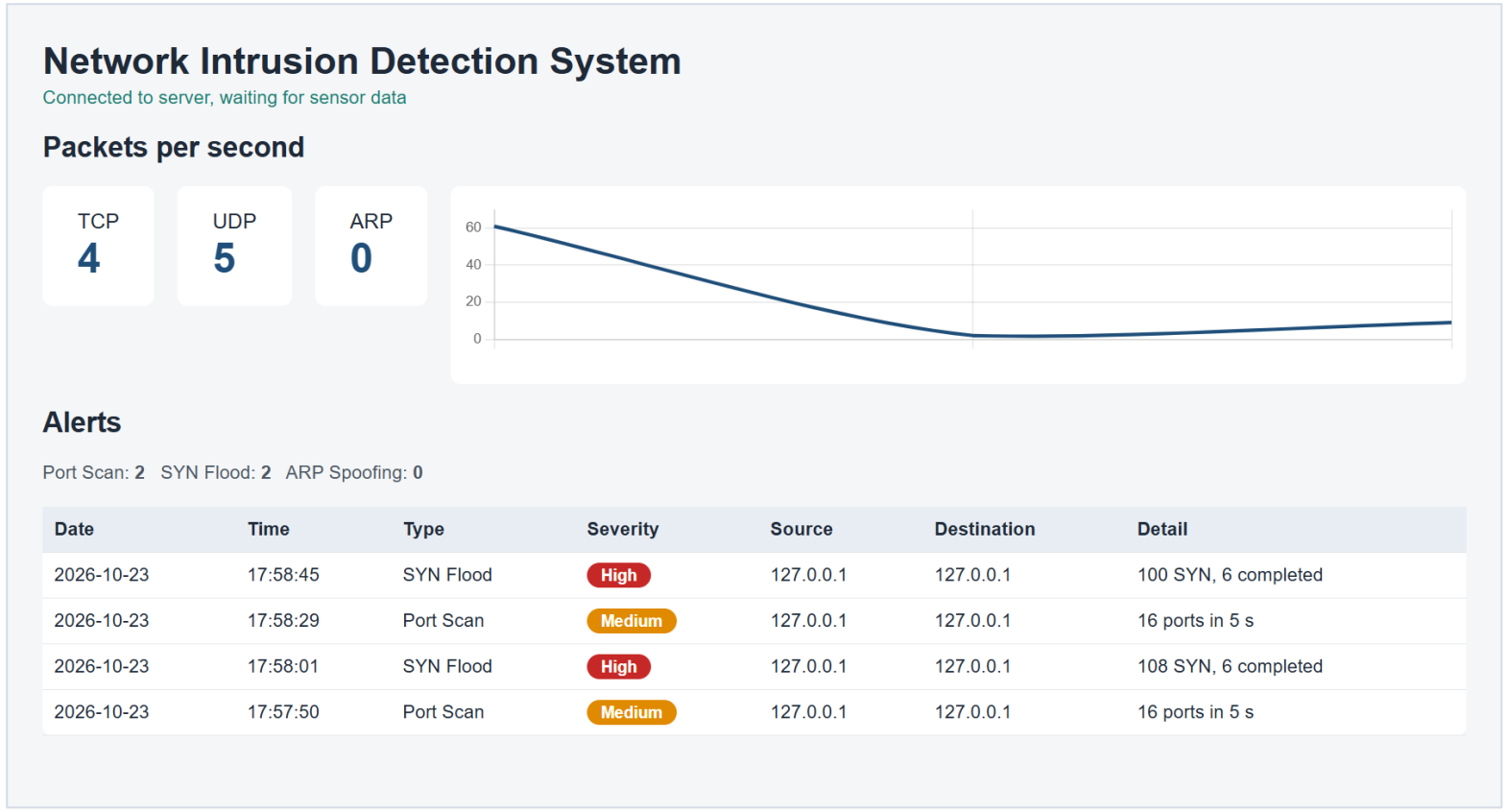
**INTEGRATION TESTS**

Real port scan and SYN flood against the host
Alert reached the dashboard and log in a few ms
No false alarm during normal browsing

## What the negative tests confirm

- Reply (SYN-ACK) packets are not a scan
- Completed handshakes are not a flood
- Genuine ARP replies raise no alarm
- Slow activity below the threshold stays silent
- A port scan is not mistaken for a flood
- A cool-down stops duplicate alerts

# Results: Live Dashboard




Dashboard after detecting a port scan and a SYN flood

**3** attacks detected

**~2 ms** detection latency

**10/10** unit tests pass

**~200** lines of code

# Conclusion and Future Scope

**A complete, working NIDS.** All objectives met: live capture, three transparent detection rules, an alert pipeline, and a real-time dashboard, verified by ten unit tests and live attacks, in about 200 lines of code.

### KEY STRENGTHS

- ✓ Every alert traceable to a threshold
- ✓ Real-time: under one second
- ✓ Small, readable, easy to extend
- ✓ Fails safely if the server is down

### FUTURE SCOPE

- More rules: ICMP flood, DNS tunnelling
- E-mail or messaging notifications
- Store alerts in SQLite, searchable
- Authenticate the sensor to the server
- Active response: firewall the attacker

# Thank You

Questions are welcome

github.com/rohanpandey436/realtime-nids

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